APPLICATION

NHA	USED ON	LTR	DATE	ECO ECP	REVISION DESCRIPTION	APVD
1213634	WSR-88D	A	251223	ECP 1038 ECO A11401	INITIAL RELEASE	SMH
		B	260414	ECP 1038 ECO A11490	UPDATE TABLE III AND OTHER REQUIREMENTS	SMH

VENDOR ITEM CONTROL DRAWING

DISTRIBUTION STATEMENT A: APPROVED FOR PUBLIC RELEASE; DISTRIBUTION IS UNLIMITED.

UNLESS OTHERWISE SPECIFIED
DIMENSIONS ARE IN INCHES
TOLERANCES ON:

MILLIMETERS	INCHES
1 PL ±	2 PL ± .02
2 PL ±	3 PL ± .010

ANGLES ± 2°

APPROVALS			
DWN	MDA 251223	RMT	LMA 251223
CHK	CLS 251223	DOC	MDW 251223
ENGR	SMH 251223	CM	
QA		ROC	

WSR-88D RADAR OPERATIONS CENTER
1200 WESTHEIMER DRIVE
NORMAN, OKLAHOMA 73069-8480

NOMENCLATURE
X14 MULTIPLIER ASSEMBLY (4A1A2A1)

SIZE	CAGE CODE	DWG NO.	REV
A	0WY55	2200737	B

SCALE: NONE

GENERAL NOTES:

1. INTERPRET THIS DRAWING IN ACCORDANCE WITH ASME Y14.100.

1.1. IDENTIFICATION OF ANY SUGGESTED SOURCE OF SUPPLY (MANUFACTURER) CONTAINED IN THIS SPECIFICATION IS NOT TO BE CONSTRUED AS A GUARANTEE OF PRESENT OR FUTURE AVAILABILITY AS A SOURCE OF SUPPLY FOR THE ITEM(S).

1.2. THE USE OF A DIFFERENT PART FROM A LISTED SUGGESTED SOURCE OF SUPPLY OR A PART FROM A NON-LISTED SUGGESTED SOURCE OF SUPPLY WHICH MEETS THE SPECIFICATIONS HEREIN SHALL OCCUR ONLY AFTER REVIEW/CONCURRENCE AND A FIRST ARTICLE QUALIFICATION BY THE WSR-88D RADAR OPERATIONS CENTER (ROC) AS WELL AS ACCEPTANCE INSPECTION BY THE NATIONAL RECONDITIONING CENTER (NRC).

2. REQUIREMENTS

2.1. ELECTRICAL:

2.1.1. CONNECTIONS: SEE TABLE I

TABLE I

PORT	DESCRIPTION	CONNECTOR	NOTES
FL1	+15 VDC IN	TURRET	
FL2	+15 VDC IN	TURRET	
E1	GROUND	TURRET	CHASSIS AT GROUND POTENTIAL
E2	GROUND	TURRET	CHASSIS AT GROUND POTENTIAL
J1	RF POWER INPUT	SMA FEMALE	50 OHM AT INPUT FREQUENCY
J2	RF OUTPUT 1	SMA FEMALE	50 OHM AT OUTPUT FREQUENCY
J3	RF OUTPUT 2	SMA FEMALE	50 OHM AT OUTPUT FREQUENCY

2.1.2. MEASURED AT 25 +/- 5°C FOLLOWING INITIAL BAKE, THERMAL SHOCK, AND BURN-IN PER SECTION 2.3.2. MEASUREMENTS MUST BE WITHIN TOLERANCE WITHIN FIVE (5) MINUTES OF POWER APPLICATION.

TABLE III ELECTRICAL REQUIREMENTS		
	RANGE	NOTES
DC VOLTAGE	+15 VDC ± 10%	
DC CURRENT DRAW	300mA MAX	
INPUT POWER	+10 ± 0.5 dBm	
INPUT FREQUENCY	57.5491MHz ± 0.0005%	
INPUT WAVEFORM	SINE WAVE	
OUTPUT FREQUENCY	805.6874MHz ± 0.0005%	BOTH OUTPUTS
OUTPUT POWER J2	+4 ± 0.5dBm	
OUTPUT POWER J3	+0.5 ± 0.5dBm	
OUTPUT WAVEFORM	SINE WAVE	
RF INPUT/OUTPUT IMPEDANCE	50 OHMs CHARACTERISTIC	RELATIVE TO OPERATIONAL FREQUENCY
PHASE NOISE GAIN	30dB MAX	IN REFERENCE TO SIGNAL SOURCE PHASE NOISE FROM 10Hz TO 1MHz OFFSET OF SOURCE FREQUENCY
MEAN PHASE NOISE GAIN	25dB MAX	IN REFERENCE TO MEAN SIGNAL SOURCE PHASE NOISE FROM 10Hz TO 1MHz OFFSET OF SOURCE FREQUENCY
VSWR	1.5:1 MAX	

2.1.3. OUTPUT SPECTRUM REQUIREMENTS

2.1.3.1. NON-HARMONIC SPURS SHALL BE NO GREATER THAN -70dBc AT EITHER OUTPUT.

2.1.3.2. 2ND ORDER AND HIGHER ORDER HARMONICS SHALL BE NO GREATER THAN -75dBc AT EITHER OUTPUT

2.2. MECHANICAL:

2.2.1. OUTLINE: PER FIGURE 1.

2.2.2. FILTER TUNING SCREWS SHOULD BE 4-40X1.125”. SHORTER SCREWS MAY BE REQUIRED IF SCREW PROTRUDES MORE THAN 3/8” FROM HOUSING WHEN TUNING IS COMPLETED.

2.2.3. ENCLOSURE MATERIAL: ENCLOSURE TO BE MADE OF 6061 ALUMINUM ALLOY PER ASTM B209.

2.2.4. CORROSION RESISTANCE: EXPOSED SURFACES TO BE COATED WITH HIGHLY CONDUCTIVE CORROSION RESISTANCE TREATMENT, SUCH AS “NICKEL-PLATING PER SAE-AMS-2404G CLASS 4, GRADE B” OR SUITABLE ALTERNATIVE FOR MATERIAL.

2.2.5. MARKING: MARK PER MIL-STD-130. LOCATE APPROXIMATELY AS SHOWN IN FIGURE DRAWINGS.

2.3. ENVIRONMENTAL:

2.3.1. INDOOR:

2.3.1.1. OPERATING TEMPERATURE: -40ºC (-40ºF) TO +85ºC (185ºF)

2.3.1.2. NON-OPERATING TEMPERATURE: -55ºC (-67ºF) TO +125ºC (257ºF)

2.3.1.3. OPERATING HUMIDITY: 15% TO 95%

2.3.1.4. NON-OPERATING HUMIDITY: 15% TO 95%

2.3.1.5. ALTITUDE: UP TO 3300 METERS ABOVE MEAN SEA LEVEL

2.3.2. STRESS TESTS:

2.3.2.1. TEMPERATURE CYCLING: 3 CYCLES FROM -55°C (-67°F) TO +125°C (257°F) PER MIL-STD 883 METHOD 1010.8 TEST CONDITION B.

2.3.2.2. STABILIZATION BAKE: MAINTAIN IN 100°C (212°F) ± 3°C ENVIRONMENT FOR TWELVE (12) HOURS

2.3.2.3. BURN-IN: DEVICE POWERED WITH RATED SUPPLY VOLTAGE FOR TWELVE (12) HOURS AT 25°C ± 5°C (77 ± 9°F)

2.3.2.4. MECHANICAL SHOCK AND VIBRATION: THE UNIT SHALL BE DESIGNED TO WITHSTAND THE VIBRATION AND SHOCK ENCOUNTERED DURING TRANSPORTATION BY AIRCRAFT, SHIPS, RAIL, AND TRUCKS. IN ADDITION, THE UNIT SHALL NOT BE DAMAGED OR LOSE PREFORMANCE CAPABILITY BY AIR TRANSPORTATION AT ALTITUDES OF UP TO 15,500 METERS (50,000 FEET) IN NON-PRESSURIZED COMPARTMENTS

2.4. RELIABILITY

2.4.1. THE MEAN-TIME BETWEEN FAILURES (MTBF) SHALL BE AT LEAST 50,000 HOURS.

2.5. PART IDENTIFICATION:

2.5.1. VENDOR IDENTIFICATION AND VENDOR PART NUMBER TO BE APPLIED BY VENDOR. PART NUMBER MAY BE APPLIED TO PACKAGING.

2.5.2. SERIAL NUMBER APPLIED AT LOCATIONS INDICATED IN FIGURE I.

3. PROCURING AGENCY:

3.1. THE PROCUREMENT AGENCY RESERVES THE RIGHT TO PERFORM ANY INSPECTION DEEMED NECESSARY TO BE ASSURED THE PARTS CONFORM TO THE SPECIFIED REQUIREMENTS.

4. THE PART NUMBER IS THE DRAWING NUMBER SUFFIXED WITH THE APPLICABLE DASH NUMBER SHOWN IN TABLE II.

5. SUGGESTED SOURCE(S) OF SUPPLY (MANUFACTURER(S)):

TBD (CAGE CODE TBD)
PART NO.: PER TABLE II

DASH NO.	ITEM	PART NO.
201	X14 MULTIPLIER ASSEMBLY	TBD

6. FIGURES AND DIAGRAMS

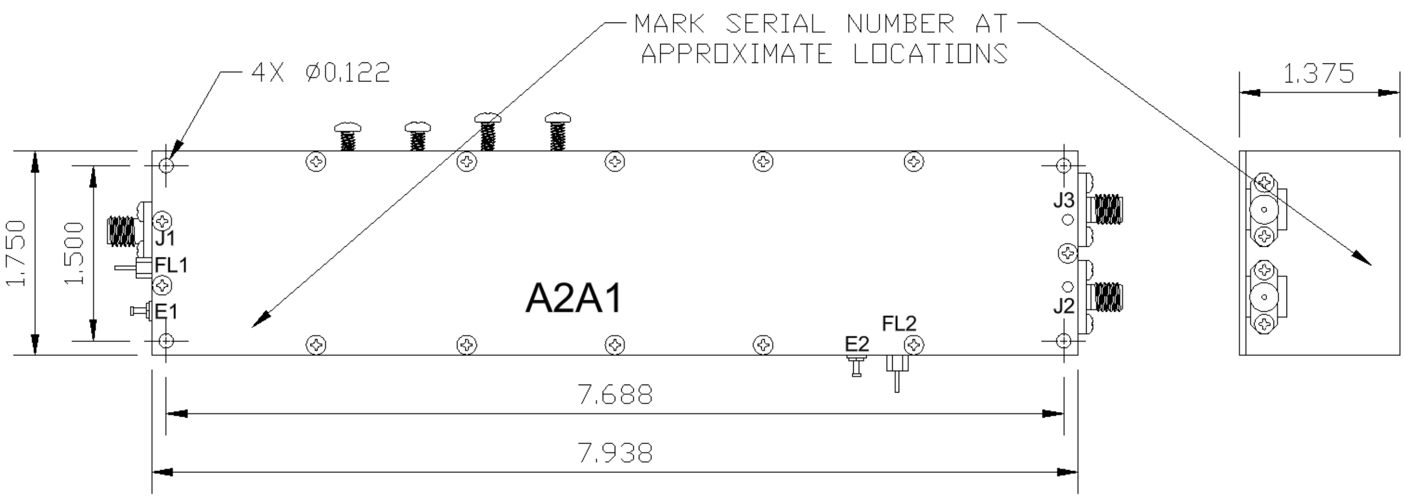

FIGURE 1